Observability-Guided LLM Query Expansion in Search: The Actionability Gap

Anonymous Author(s)

Abstract

LLM query expansion (QE) helps some queries and harms others, which motivates applying it selectively. A selective controller needs two predictions: whether retrieval is failing (*detection*) and whether a specific expansion will help (*action utility*). We measure both on the same queries from qrel-free retrieval telemetry, on TripClick and BEIR Natural Questions, under BM25 and a dense retriever. On the queries the system expands, detection AUC is 0.62–0.70, no better than a standard unsupervised predictor (NQC), while predicting whether the expansion harms retrieval reaches only 0.55–0.61: a paired-significant gap in three of four settings. The gap is largely structural. Expansion leaves 84–94% of failing queries still failing, and among queries whose nDCG the expansion changed, the gap persists only on dense NQ. The policy picture matches. A perfect failure detector would recover only 42% (TripClick) and 33% (NQ) of an oracle selector's gain; the rest lies in healthy queries that expansion sometimes helps and sometimes harms. Our failure-selective controller halves generator calls, and on NQ it significantly reduces the harm of unconditional expansion, but no policy measurably beats BM25 alone.

1 Introduction

LLMs can rewrite a query into pseudo-documents, paraphrases, or disambiguating context, and conditioning the rewrite on retrieved evidence improves on query-only rewriting [12]. Applied unconditionally, though, LLM QE harms a measurable fraction of queries [1, 20]. Selective expansion is the natural response, and it needs two predictions:

$$\underbrace{P(\text{failure} \mid x)}_{\text{detection}} \qquad \text{vs.} \qquad \underbrace{P(\Delta\text{nDCG@10} < 0 \mid x, a)}_{\text{action utility (harm)}}$$

Detection asks whether the current ranking is already bad. Action utility asks whether a *specific* expansion a will make it worse. That the two differ is not new: selective-QE work predicts expansion benefit directly [2, 7], and uplift modelling shows that targeting high-risk cases is not the same as targeting cases a treatment helps [3, 8]. What is missing for LLM QE is a measurement of both predictions on the same queries, and of what each one is worth to a policy. We use *observability* for qrel-free telemetry a search system records about its own behaviour (score distributions, coverage, ranking stability, reranker agreement), the signal a deployed controller can actually see.

Contributions. (C1) A paired, same-query measurement of detection and action-utility AUC on two corpora under BM25 and dense retrieval, with the label's tie rule, a standard QPP baseline, and a combined pre+post-action model reported alongside. **(C2)** A decomposition of selective-QE headroom: how much of an oracle selector's gain perfect failure detection alone would recover, and why the remainder needs action-utility prediction. **(C3)** A policy evaluation with minimum detectable effects: failure-selective QE halves generator calls and, on NQ, significantly reduces unconditional QE's harm, but no policy detectably beats BM25.

2 Related Work

Selective QE and query performance prediction. Pre- and post-retrieval QPP (Clarity [6], WIG [23], NQC [17]) predicts query difficulty. Selective QE goes further: Cronen-Townsend et al. [7] decide whether to expand by comparing the clarity of the expanded and unexpanded rankings, i.e. they predict expansion benefit directly; Amati et al. [2] gate expansion on difficulty predictors; Lv and Zhai [13] predict a per-query feedback weight; and Collins-Thompson [5] trades expansion reward against risk. These works evaluate the gated system in aggregate; they do not report detection and benefit prediction as two AUCs on the same queries, nor decompose the headroom between them.

LLM QE and its failures. Query2doc [19], MuGI [22], and CSQE [12] expand unconditionally. Weller et al. [20] find generative expansions often hurt strong retrievers; Abe et al. [1] trace failures to unfamiliar or ambiguous queries. Kotte [10] gates prompt-only rewriting for dense retrievers (gate AUC 0.593, no reliable gain over never rewriting); Adaptive-RAG [9] routes on a question-complexity classifier. Neither separates the gate's detection accuracy from its benefit prediction.

Uplift modelling. Predicting an outcome is not predicting a treatment's effect on it [8]; targeting the customers most likely to churn can be ineffective [3]. Our setting has an advantage marketing lacks: both outcomes (expanded and unexpanded) are observed for every query offline, so the uplift label is exact up to relevance-label noise.

3 Setup

Data. TripClick [15]: 1.52M health documents; the TAIL query bucket's validation and test splits (1,175 queries each), with the click-derived relevance labels TripClick releases. BEIR NQ [11, 18]: 2.68M passages, human-annotated relevance; we carve a fixed-seed split (800 validation) and use a disjoint 2,052-query test set, run once under a scope committed before any of its data was touched. An earlier n=600 NQ pilot is reported only in the ledger below.

Pipeline. BM25 over SQLite FTS5 (Porter stemming, title weight selected on validation). From pre-action telemetry a deterministic rule picks one of four expansion operators or none; it picks none for 1.5% (TripClick) and 1.9% (NQ) of queries. The rest form the *acted-upon* set, on which every AUC in Table 1 is computed. The generator (`qwen2.5:3b`, 4-bit, greedy, ≤ 2 calls per query) writes a corpus-grounded expansion from the top retrieved evidence; the expanded query is retrieved and, on TripClick and in both dense settings, reranked with a MiniLM cross-encoder [16] (the NQ BM25 run does not rerank). The dense setting replaces first-stage retrieval with `bge-small-en-v1.5` [21] over the full collections.

Labels. *Failure*: the first-stage top 10 contains no relevant document (nDCG@10 = 0); 45% of TripClick and 49% of NQ test queries under BM25. *Action utility*: Δ = nDCG@10(expanded) − nDCG@10(original), with *harmed* ($\Delta < 0$) as the positive class and ties ($\Delta = 0$) counted as not harmed. Ties are 48% (TripClick) and 69% (NQ) of acted-upon queries, and harm 33% and 17%. Harm is impossible on a failing query, since its nDCG is already 0.

Models and features. Detection: logistic regression on 8 pre-action features (top score, score margin, score entropy, query-term coverage, mean rare-term IDF, entity overlap, result coherence, ranking stability under a small query perturbation); 6 under dense retrieval (no IDF, no stability). Action utility: 10 post-action features (gains in score margin, coverage and coherence; stability; rewrite drift; reranker–first-stage disagreement; top-k overlap with the original; reranker top score and margin; query-length ratio), fit by linear regression on Δ under BM25 and as a logistic harm classifier under dense retrieval (9 features, no stability). Every model and threshold is fit on validation and scored once on test; significance is a paired bootstrap over queries (10^4 resamples, seed 42).

Policies. *BM25*: never expand. *Static QE*: always expand; TripClick deploys a reciprocal-rank fusion of the original and expanded rankings (no fused ranking was stored for NQ, whose arms use the expanded ranking itself). *Failure-selective QE*: expand iff $P(\text{fail} \mid x) > \tau$, τ the median validation prediction. *Closed-loop QE* (TripClick only): expand, then accept the fused ranking, roll back, or re-plan once based on the action-utility score, with validation-calibrated thresholds.

BM25 baseline. Our first, unstemmed BM25 scored nDCG@10 0.228 on TripClick TAIL test, below the published 0.267 [15]. Re-selected on validation only (validation 0.235 → 0.327), BM25 alone (no reranker) scores 0.315 on test.

Test-access ledger. TripClick TAIL test was used during development before our data-use protocol existed: an early verifier and its thresholds were calibrated on it, two apparatus defects were found there (the reranked order was not wired into the accepted ranking; the verifier judged a different ranking than the one accepted), and the weak 0.228 BM25 was observed there. We then froze the protocol, refit every model and threshold on validation, checked BM25 once (0.315), and ran one frozen evaluation that produced every TripClick test episode used here, plus a 150-query dense pilot and one full dense run. All later analyses, including this revision's, recombine those frozen records under pre-specified rules without refitting on test. NQ test was touched three times, each under a written pre-commitment: the n=600 pilot (detection 0.638 [0.593, 0.681], action utility 0.522 [0.442, 0.602]), the confirmatory 2,052-query run, and its dense rerun; two later BM25 re-runs on the same queries recomputed telemetry and score lists only. Scripts and pre-commitments are in the anonymized repository: https://anonymous.4open.science/r/agentic-query-expansion-8939/.

4 Result 1: Detection vs. Action Utility

Under the pre-committed harm label, detection beats action utility on the same queries in three of four settings (Table 1). On the full test populations, detection AUC is 0.649, 0.680, 0.638, and 0.688. Dense TripClick is the exception; there the bi-encoder is weaker than BM25 (nDCG@10 0.217 vs. 0.315), while on NQ it is stronger (0.500 vs. 0.308).

Detection is ordinary QPP. The unsupervised NQC predictor over the top-100 BM25 scores [17] (normalized by the top-k mean, as FTS5 exposes no corpus score) matches our detector on TripClick (0.647 vs. 0.649; paired difference +0.003 [−0.030, +0.035]) and beats it on NQ (0.721 vs. 0.680; −0.041 [−0.064, −0.017]). WIG [23] is weaker (0.588 and 0.651). Detection is not the novel part; it is what selective QE already has.

Not a feature-set artifact. A logistic harm model on pre-action *plus* post-action features (18 under BM25, 15 dense) still trails detection on TripClick BM25 (0.603; difference +0.040 [+0.003, +0.078]) and dense NQ (0.625; +0.070 [+0.033, +0.107]), and does not differ from the post-action model alone on TripClick (+0.020 [−0.013, +0.053]). Pre-action features alone predict harm about as well as post-action ones on TripClick (0.592 vs. 0.582). A logistic refit of the BM25 action-utility model gives 0.563 (+0.011 [−0.008, +0.029] over the linear one).

Not label sparsity. Most TAIL queries have one relevant document, so failure might track label sparsity. The relevant-document count alone predicts failure at only 0.570 (TripClick) and 0.521 (NQ). Within count strata, detection AUC is 0.644 and 0.677 (TripClick; size-weighted 0.656) and 0.687 and 0.659 (NQ), and the detector stays significant when the count is controlled for.

The gap is largely structural. Expansion rarely rescues a failing query: 84%, 93%, 94%, and 84% of acted-upon failing queries keep nDCG@10 = 0 (ties). Failing queries are therefore mostly "not harmed", so the two labels are coupled, and the harm AUC compares a detector against a label that is itself largely a failure indicator. Restricted to queries whose nDCG the expansion changed (last two columns), the BM25 gaps vanish (point estimates reverse, CIs span zero) because detection itself drops to 0.53–0.58 there; only dense NQ keeps a significant gap (+0.122). Predicting the *direction* of a change is weak everywhere (0.51–0.62). It is better for large effects: on the half of changed queries with the largest $|\Delta|$, it reaches 0.644 and 0.711 under BM25, though only 0.551 and 0.526 under dense retrieval. Under BM25, the action-utility score's errors are therefore concentrated in small changes.

5 Result 2: What Is Each Prediction Worth?

Policies. On TripClick, no policy differs from BM25 or from any other; every pairwise CI spans zero (Table 2). These nulls are bounded: with MDEs of 0.007–0.011 nDCG@10, effects smaller than that could exist undetected. On NQ, where the expanded ranking is used unfused, unconditional QE significantly harms (−0.022). Failure-selective QE recovers most of that loss (+0.015 [+0.008, +0.022] over Static QE) and is not distinguishable from BM25. Failure-selective QE routes 49% (TripClick) and 51% (NQ) of queries to the generator, halving calls relative to Static QE. The closed-loop agent costs 24% more calls (1.22 vs. 0.98 per query; Static QE skips the 1.5% of queries the planner leaves unexpanded) and 36% more latency than Static QE, for no measurable gain.

Headroom decomposition. The oracle selector expands a query iff that helps; it is non-negative by construction and is an upper bound, not a significance result. A *perfect detector* expands exactly the failing queries. Because QE can only help a failing query

Table 1: Detection vs. action-utility (harm) AUC on each setting's acted-upon queries, with 95% CIs. "Diff." is the paired bootstrap of their difference on the same queries, so it equals the difference of the two columns. The last two columns use only queries whose nDCG@10 the expansion changed ($\Delta \neq 0$): action utility as helped-vs.-harmed, and the paired difference (post hoc, not pre-committed).

Setting	n	Detection	Action utility	Diff.	Direction, $\Delta \neq 0$	Diff., $\Delta \neq 0$
TripClick, BM25	1157	0.643 [0.612, 0.674]	0.552 [0.518, 0.586]	+0.091 [+0.045, +0.136]	0.572 [0.524, 0.620]	−0.044 [−0.127, +0.039]
NQ, BM25	2013	0.675 [0.652, 0.698]	0.556 [0.517, 0.594]	+0.119 [+0.076, +0.162]	0.624 [0.581, 0.667]	−0.041 [−0.122, +0.036]
TripClick, dense	1050	0.621 [0.587, 0.655]	0.599 [0.555, 0.642]	+0.022 [−0.028, +0.072]	0.558 [0.497, 0.617]	+0.043 [−0.063, +0.145]
NQ, dense	1932	0.695 [0.669, 0.720]	0.611 [0.583, 0.639]	+0.084 [+0.045, +0.123]	0.512 [0.471, 0.552]	+0.122 [+0.047, +0.197]

Table 2: nDCG@10 gain over BM25 on the full test sets, with 95% CIs and, for real policies, the minimum detectable effect at 80% power (MDE). Retrospective rows use test labels and are upper bounds, not policies. NQ arms use the unfused expanded ranking.

	TripClick (n=1175)	NQ (n=2052)
BM25 nDCG@10	0.3152	0.3080
Static QE	+0.0007 [−0.007, +0.008]	−0.0217 [−0.032, −0.012]
Failure-selective QE	+0.0027 [−0.002, +0.008]	−0.0067 [−0.014, +0.000]
Closed-loop QE	+0.0009 [−0.007, +0.009]	–
MDE (real policies)	0.007–0.011	0.010–0.015
Perfect detector	+0.0103 [+0.007, +0.014]	+0.0135 [+0.010, +0.017]
Oracle selector	+0.0246 [+0.020, +0.030]	+0.0415 [+0.036, +0.047]

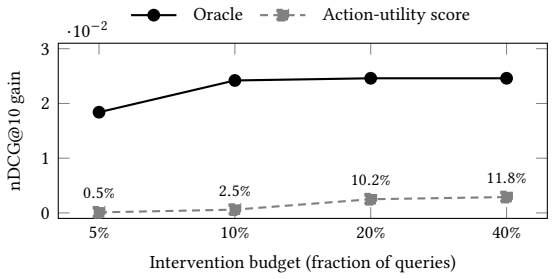


Figure 1: Uplift (Qini-style [14]) curve on TripClick test: gain over BM25 when expanding the top-k queries ranked by the oracle or by the action-utility score. Labels give the score's share of the oracle's gain (mean 6% across budgets).

(TripClick +0.023 on failing vs. −0.018 on healthy queries; NQ +0.028 vs. −0.069), perfect detection is safe, but it recovers only 42% (TripClick) and 33% (NQ) of the oracle's gain: most failing queries are not rescued. The remaining 58% and 67% lies in healthy queries that expansion sometimes helps and sometimes harms, which only an action-utility predictor can separate. Our detector captures part of the first share: its routed queries gain +0.0055 [−0.005, +0.016] on TripClick, against −0.0040 for the rest.

Heterogeneity is real, and partly predictable. Ranking queries by the action-utility score captures at most 12% of the oracle's gain (Figure 1). Yet the score is not noise. In a best-linear-predictor test [4] of Δ on the validation-fit score under BM25, the slope is +1.09 [+0.54, +1.67] on TripClick and +1.21 [+0.96, +1.44] on NQ, and the top score quartile's mean Δ exceeds the bottom quartile's by +0.093 and +0.120 (both CIs exclude zero). Treatment effects vary across queries, and post-action telemetry tracks them weakly. The oracle's magnitude is split-sensitive on TripClick (validation +0.053 [0.045, 0.062] vs. test +0.025 [0.020, 0.030]) but not on NQ (+0.040 [0.031, 0.049] vs. +0.042 [0.036, 0.047]), so we read it as an order of magnitude, not a bound to target.

6 Discussion and Limitations

Why action utility is harder. Both outcomes are observed offline, so the difficulty is not counterfactual missingness. Detection describes a ranking that already exists. Action utility depends on what a generator writes from evidence it did not choose; the output is deterministic given its input (greedy decoding) but varies sharply across queries, and its effect on a single relevant document's rank is small and discrete. That fits the evidence above: direction is predictable for large changes and close to chance for small ones.

Limitations. One small (3B) generator under a two-call budget; a stronger one may change both the rescue rate and the policy results. Two corpora; under dense retrieval, retriever strength and domain are confounded. TripClick's relevance labels are click-derived and sparse, which is one reason we report NQ's human labels in parallel. The policy comparisons on NQ use unfused rankings and lack a closed-loop arm. The tie analysis in Table 1 was not pre-committed.

7 Conclusion

Detecting that retrieval has failed is ordinary QPP; our detector is no better than NQC. Knowing that an expansion will fix it is a separate problem. Expansion rarely rescues a failing query, so even perfect failure detection recovers only a third to two-fifths of selective QE's headroom. The rest requires predicting whether a specific rewrite helps a query that is already working, which our telemetry does only weakly. In practice, failure-selective QE halves generator cost and avoids most of unconditional QE's harm, but it does not by itself deliver gains over BM25. Selective-QE work should therefore report action utility on the same queries as detection, together with its tie rule, rather than treating a failure detector as a proxy for benefit.

Ethical Considerations

All data used (TripClick, BEIR Natural Questions) are existing research benchmarks, used under their licenses; TripClick is used for non-commercial research, and no model or per-query output fit on it is released. TripClick's click logs are aggregate, de-identified interactions, and we make no attempt to re-identify searchers. No

human-subjects data was collected. The generator is a small, locally hosted model used only to rewrite queries within a bounded action space. Our central finding cautions against deploying generative interventions on the strength of a failure detector alone.

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